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Series 9: Abstract Interpretation I

Methods

Construction of CFG (Control Flow Graph)

- **Inductive method**: Define a recursive function `CFG(P, ip, rp)` that constructs the control flow graph for a program `P`, with an entry point `ip` and a return point `rp`.
- **Construction rules**:
  - **Assignment**: `CFG(x:=e, ip, rp)` adds a transition labeled `x:=e` from `ip` to `rp`.
  - **Sequence**: `CFG(e1; e2, ip, rp)` creates an intermediate point `e1` and recursively calls `CFG(e1, ip, e1)` and `CFG(e2, e1, rp)`.
  - **Conditional**: `CFG(if c then e1 else e2, ip, rp)` creates two new points `e1` and `e2`, adds transitions from `ip` to `e1` (label `c`) and from `ip` to `e2` (label `¬c`), then recursively calls `CFG(e1, e1, rp)` and `CFG(e2, e2, rp)`.
  - **Loop**: `CFG(while c do e1 done, ip, rp)` creates a point `e1`, adds a transition from `ip` to `e1` (label `c`) and one from `ip` to `rp` (label `¬c`), then calls `CFG(e1, e1, ip)`.

Collecting Semantics

- **System of equations**: Associate with each control point `s` a variable `V(s)` representing the set of possible memory states at that point.
- **Equations**: Deduce equations from the CFG structure, using union operations and filtering by conditions.
- **Entry point**: Introduce an initial state `st` to model parameter initialization.

Iterative Resolution of the Equation System

- **Iteration to fixed point**: Initialize `V(s0)` with the initial state, then recursively apply equations until stabilization (or divergence).
- **Termination analysis**: Observe value evolution to conclude about program behavior (termination, infinite loop, error).

Abstraction by Intervals

- **Abstract domain**: Replace concrete sets of values with intervals (e.g., `[2, 6]` for `{2,3,4,5,6}`).
- **Abstract operations**: Adapt operations (assignment, union, intersection) to work on intervals.
- **Goal**: Simplify analysis by losing precision, but guaranteeing termination and capturing general properties.

Abstraction by Signs

- **Abstract domain**: Represent sets of values by their sign: `0`, `0`, `<0`, `>0`, `=0`, `0`, (no value), (all values).
- **Abstract operations**: Define operations on signs (e.g., addition, subtraction, comparison).
- **Goal**: Even coarser but faster analysis, to infer simple properties about variables.

Reminder

- **CFG**: Graphical representation of program control flow. Nodes are control points, arcs are transitions labeled by conditions or assignments.
- **Collecting semantics**: Semantics that associates with each control point the set of all possible memory states during execution.

- **Fixed point**: In abstract interpretation, we look for a fixed point of the equation system (a state where `V(s)` no longer changes after an iteration). The iterative algorithm converges to this fixed point (under certain conditions).
- **Abstraction**: Function that maps a concrete set (e.g., `{2,3,4}`) to an element of an abstract domain (e.g., interval `[2,4]` or sign `>0`). It must be safe (over-approximative).
- **Abstract operations**: Must over-approximate concrete operations. For example, interval addition `[a,b] + [c,d] = [a+c, b+d]`.

Illustrative Examples

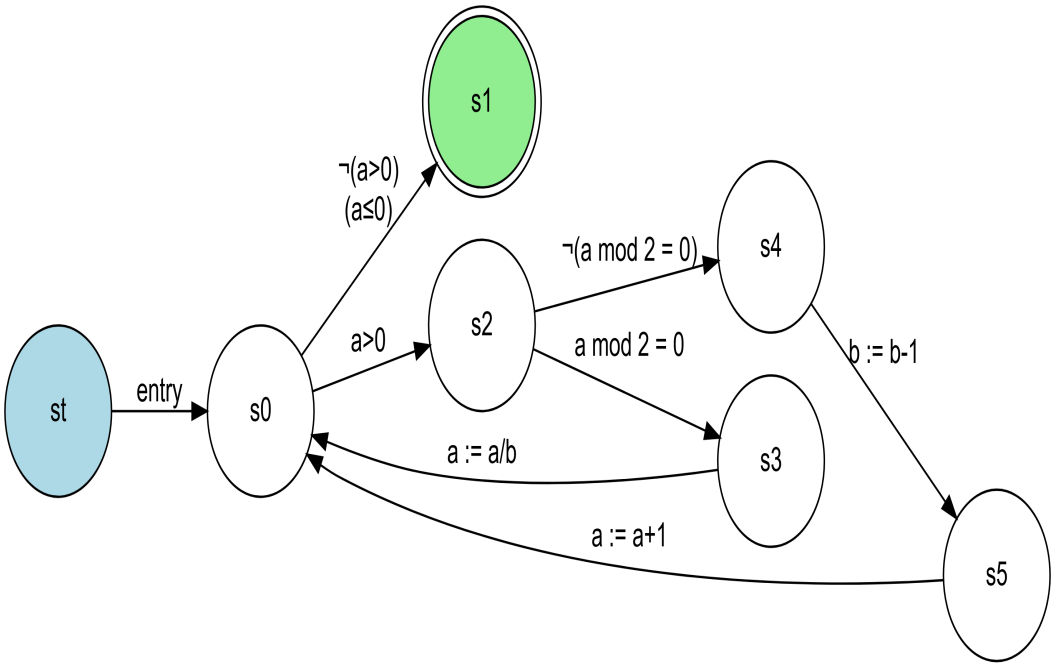
Example 1: CFG Construction for `sillyprogram`

Context: Program `sillyprogram(a,b) = while (a > 0) do if a mod 2 = 0 then a := a/b else b := b-1; a := a+1 endif done`. We want to construct its CFG.

Application: Apply inductive rules.

1. **Loop while**: `CFG(while (a>0) do ... done, ip, rp)`
   - Creates a new point `e1` for the loop body.
   - Adds a transition from `ip` to `e1` labeled `a>0`.
   - Adds a transition from `ip` to `rp` labeled `¬(a>0)` (i.e., `a<=0`).
   - Recursive call: `CFG(body, e1, ip)` where `body` is the `if ... endif` instruction.
2. **Conditional if**: `CFG(if a mod 2 = 0 then a:=a/b else b:=b-1; a:=a+1, e1, ip)`
   - Creates two points `e2` (for then) and `e3` (for else).
   - Adds a transition from `e1` to `e2` labeled `a mod 2 = 0`.
   - Adds a transition from `e1` to `e3` labeled `¬(a mod 2 = 0)`.
   - Recursive calls:
     - `CFG(a:=a/b, e2, ip)`
     - `CFG(b:=b-1; a:=a+1, e3, ip)`
3. **Sequence**: `CFG(b:=b-1; a:=a+1, e3, ip)`
   - Creates an intermediate point `e4`.
   - Recursive calls:
     - `CFG(b:=b-1, e3, e4)`
     - `CFG(a:=a+1, e4, ip)`

The resulting CFG has control points `st` (initial), `s0` (loop start), `s1` (loop exit), `s2` (if start), `s3` (then branch), `s4` (else branch), `s5` (after `b:=b-1`). Transitions are labeled as described.



Example 2: Collecting Semantics for `sillyprogram`

Context: We have the CFG of `sillyprogram`. We want to write the collecting semantics equation system.

Application: Define `V(s)` for each control point `s`.

$$
\begin{aligned}
V(st) &= \top \quad \text{(unspecified initial state)} \\
V(s0) &= V(st)[a := a_0, b := b_0] \cup V(s3)[a := a/b] \cup V(s5)[a := a + 1] \\
V(s1) &= V(s0) \cap [a \leq 0] \\
V(s2) &= V(s0) \cap [a > 0] \\
V(s3) &= V(s2) \cap [a \mod 2 = 0] \\
V(s4) &= V(s2) \cap [a \mod 2 \neq 0] \\
V(s5) &= V(s4)[b := b - 1]
\end{aligned}
$$

Here, `[cond]` denotes the set of states satisfying condition `cond`. The operation `V(s)[x:=expr]` means "take all states from `V(s)`, evaluate `expr` in each state, and assign the resulting value to `x`".

Example 3: Concrete Resolution for `a=5, b=3`

Context: Solve the equation system for initial values `a=5, b=3`.

Application: Iterate manually.

- Initialization: `V(s0) = {a:{5}, b:{3}}`
- First iteration (loop):
  - Via `s2` (since `a>0`): `V(s2) = {a:{5}, b:{3}}`
  - Via `s4` (since `a mod 2  0`): `V(s4) = {a:{5}, b:{3}}`
  - Via `s5`: `V(s5) = {a:{5}, b:{2}}`
  - Return to `s0`: `V(s0) = {a:{5}, b:{3}}  {a:{6}, b:{2}} = {a:{5,6}, b:{2,3}}`
- Second iteration: Recompute with the new `V(s0)`. We get `V(s0) = {a:{2,3,5,6}, b:{1,2,3}}`.
- Third iteration: `V(s0) = {a:{0,1,2,3,5,6}, b:{0,1,2,3}}`.
- Fourth iteration: `V(s0) = {a:{0,1,2,3,5,6}, b:{-1,0,1,2,3}}`.
- Fifth iteration: Sets continue to expand (appearance of negative values). We conclude that the loop does not terminate and values diverge.

Example 4: Abstraction by Intervals

Context: Take the same system, but abstract value sets with intervals.

Application: Re-express equations with intervals.

- Initialization: `V(s0) = {a:[5,5], b:[3,3]}`
- First iteration:
  - `V(s2) = {a:[5,5], b:[3,3]}`
  - `V(s4) = {a:[5,5], b:[3,3]}`
  - `V(s5) = {a:[5,5], b:[2,2]}`
  - `V(s0) = {a:[5,5], b:[3,3]}  {a:[6,6], b:[2,2]} = {a:[5,6], b:[2,3]}`
- Second iteration: We get `V(s0) = {a:[2,6], b:[1,3]}`.
- Third iteration: `V(s0) = {a:[0,6], b:[0,3]}`.
- Fourth iteration: `V(s0) = {a:[0,6], b:[-1,3]}`.
- Fifth iteration: `V(s0) = {a:[-6,6], b:[-2,3]}` (approximating appeared negative values). The interval for `a` expands toward negatives, and we can conjecture it will become `(-ϖ, +ϖ)` after enough iterations.

Example 5: Abstraction by Signs

Context: Now use sign abstraction.

Application: Translate values to signs.

- Initialization: `V(s0) = {a:>0, b:>0}` (since 5>0, 3>0)
- First iteration:
  - `V(s2) = {a:>0, b:>0}`
  - `V(s4) = {a:>0, b:>0}`
  - `V(s5) = {a:>0, b: 0}` (since b-1 can be 2>0, but we over-approximate by 0)
  - Return to `s0`: Union of signs: `{a:>0, b:>0}` `{a:>0, b: 0} = {a:>0, b: }` (since >0 0 = for b)
- Second iteration: With `b: `, abstract calculations will yield very broad signs. Quickly, we get `V(s0) = {a: , b: }`, meaning any value is possible for `a` and `b`. This indicates that sign analysis is too coarse for this program, but it allows concluding that the program may not terminate (since we cannot guarantee that `a` becomes 0).

Synthesis

Abstract interpretation is a static analysis technique that approximates program behavior without concretely executing it. First construct a CFG, then define a collecting semantics equation system. Iterative resolution of this system may not terminate on complex programs. To guarantee termination, we use abstract domains (intervals, signs) that simplify data and allow computing a fixed point in a finite number of steps. However, abstraction causes loss of precision: we can obtain safe results but sometimes too broad.